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$$\sum_{n=2}^{\infty} \frac{2^{2n}}{2n!}$$

$\Rightarrow$ D'Alembert

$$\frac{\frac{2^{2n+2}}{(2n+2)!}}{\frac{2^{2n}}{2n!}} = \frac{2^{2n+2}}{(2n+2)!} \cdot \frac{2n!}{2^{2n}}$$

$$= \frac{4}{(2n+2)(2n+1)}$$

$$\Rightarrow \lim_{n \to +\infty} \frac{4}{(2n+2)(2n+1)} = 0 < 1 \Rightarrow \text{convergent}$$

References